

POORAB GUPTA

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PROFILE SUMMARY

BCA graduate specializing in Generative AI and agentic systems, with hands-on experience building multi-step LLM applications using LangChain, LangGraph, and RAG pipelines. Solid backend foundation (FastAPI, Docker, AWS) and a complementary background in network security and vulnerability assessment. Builds and ships independent AI projects end-to-end, from architecture through deployment.

TECHNICAL SKILLS

GenAI & Agentic AI: LangGraph, LangChain, RAG Pipelines, Vector Embeddings, Prompt Engineering, Tool-Calling, Gemini API

Vector Databases: Pinecone, Chroma, FAISS

Backend & Cloud: Python, FastAPI, Flask, REST APIs, Docker, AWS EC2, Nginx, Git

Cybersecurity: Network Protocols, Vulnerability Assessment, Cryptography, Nmap, Burp Suite, Metasploit, Wireshark

Languages & Platforms: Python, C, C++, Java, Linux (Kali), VS Code

PROJECTS

Aria — Personal Voice Assistant | *Python, LangGraph, Gemini API, faster-whisper, PyQt6*

- Built a Jarvis-style Windows voice assistant powered by a LangGraph ReAct agent (Gemini 2.0 Flash) orchestrating 15+ tools across OS control, media, and third-party APIs.
- Implemented a fully offline speech pipeline (faster-whisper STT, pyttsx3 TTS) with custom energy-based VAD and wake-word detection, eliminating per-request API costs.
- Designed a dynamic app-discovery system using Windows Registry and Start Menu scanning with fuzzy matching, removing the need for manual configuration.
- Architected a thread-safe, 3-mode operating system (Normal/Mute/DND) paired with a custom PyQt6 status overlay for real-time agent state.
- Applied security-first practices, including git history remediation after credential exposure and protected-process safeguards for destructive OS actions.

YouTube RAG Chatbot | *Chrome Extension (Manifest V3), FastAPI, LangChain, Pinecone*

- Built a Manifest V3 Chrome Extension answering questions from YouTube video transcripts using Retrieval-Augmented Generation.
- Developed a FastAPI backend using the YouTube Transcript API, LangChain, and Gemini embeddings, with Pinecone for vector storage and retrieval.

Cyber.AI — NLP-Based Vulnerability Analysis Tool | *Python, NLP, NVD API, Vulners API*

- Built a Python NLP tool that retrieves and summarizes CVE vulnerability data for faster triage.
- Integrated the NVD and Vulners APIs for real-time vulnerability data retrieval and classification.

Taskflow | *Flask, Docker, Nginx, AWS EC2*

- Dockerized a Flask To-Do REST API with an Nginx reverse proxy, deployed on AWS EC2; evolved from a semester project into a fully containerized deployment.

EXPERIENCE & PRACTICAL TRAINING

Cybersecurity Intern — Wayspire *July 2024 – August 2024*

- Assisted in vulnerability assessments and penetration testing using Nmap, Burp Suite, and Metasploit.
- Prepared security reports and remediation recommendations for identified weaknesses.

Offline Cybersecurity Bootcamp — CDAC Noida *January 2025*

- Hands-on training in network security, ethical hacking, and digital forensics, including live labs on web app security, cryptography, and threat hunting.

Hands-On Security Labs

- Hack The Box (Beginner & Intermediate, Web & Network) · OverTheWire Bandit & Natas · PortSwigger Web Security Academy (OWASP Top 10).

EDUCATION

Bachelor of Computer Applications (BCA) — IIIT, Noida *CGPA: 8.12*

Senior Secondary (CBSE, Science) — Our Lady of Fatima School, Aligarh *87%*

Secondary (CBSE) — Our Lady of Fatima School, Aligarh *97%*

ACHIEVEMENTS & LEADERSHIP

- Winner, Game Development Competition.
- Recipient, UP State Scholarship for college.
- Co-founded IIIT's coding club, Zen-Coders, and mentored 50+ peers and juniors in core programming concepts through lecture sessions and presentations.